

CSES Problem Set

Counting Rooms

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You are given a map of a building, and your task is to count the number of its rooms. The size of the map is $n \times m$ squares, and each square is either floor or wall. You can walk left, right, up, and down through the floor squares.

Input

The first input line has two integers n and m : the height and width of the map.

Then there are n lines of m characters describing the map. Each character is either `.` (floor) or `#` (wall).

Output

Print one integer: the number of rooms.

Constraints

- $1 \leq n, m \leq 1000$

Example

Input:

```
5 8
#####
#..#...#
####.#.#
#..#...#
#####
```

Output:

```
3
```